ENGINEERING APPROACH

Engineering decisions made *before they become expensive.*

Good engineering reduces uncertainty one decision at a time. We connect architecture, implementation, integration and validation so the next step is based on evidence rather than guesswork.

The process adapts to the project: sometimes the first job is to understand an inherited system; sometimes it is to make a prototype measurable.

01 ENGINEERING PROCESS

From uncertainty *to evidence.*

01

Understand

Requirements, constraints, existing hardware and software, failure modes, available documentation and project risks.

02

Architect

System boundaries, hardware and software interfaces, communications, control strategy, test strategy and documentation structure.

03

Build

Firmware, electronics, embedded systems, controls, automation and prototype integration shaped around the decisions that matter.

04

Validate

Measurements, repeatable tests, data analysis, debugging, technical reporting and handoff documentation.

02 HOW IT APPLIES

The same discipline *across the system.*

Embedded systems

Firmware architecture, MCU integration, communication interfaces and control implementation.

Power electronics

Prototype bring-up, measurements, switching behavior and design iteration guided by bench evidence.

Test automation

Repeatable validation, instrumentation, data collection and analysis that make results easier to review.

System recovery

Understanding inherited systems, finding failure points and stabilizing incomplete work.

03 ENGINEERING PRINCIPLE

Make the critical decision *while it is still cheap to change.*

Early architecture and validation expose expensive assumptions while there is still room to change the design, the interface or the test method.

04 WORKING METHOD

Leave a system *another engineer can use.*

The work is documented at the level the project needs: requirements and assumptions, interfaces, test conditions, measurements, decisions and next actions.

The result is not a presentation about the system. It is a clearer system, a traceable technical record and a practical path to the next decision.

START A TECHNICAL DISCUSSION

Bring the *uncertainty.*

Describe the project, its current stage and the technical question that needs a better answer. Secure file exchange can follow the initial discussion.

START A TECHNICAL DISCUSSION →

Or write directly: info@herdersystemen.com